

Calculus II

Tangents and curve length

Todor Milev

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Outline

- 1 Tangents to Curves
 - Tangents to Polar Curves

- 2 Arc Length
 - Arc Length in Polar Coordinates

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Tangents

Let C be the curve $C : \begin{cases} x = f(t) \\ y = g(t) \end{cases}, t \in [a, b]$.

Definition

Suppose $f'(t)$ and $g'(t)$ are not simultaneously equal to 0.

- We define $(f'(t), g'(t))$ to be the *tangent vector* to C at t .
- We define the line passing through $(f(t), g(t))$ with direction vector equal to the tangent vector to be *tangent line* to C at t . In other words, the tangent line has equation

$$(x - f(t))g'(t) = (y - g(t))f'(t) \quad .$$

- We say that the tangent to C at t is vertical if $f'(t) = 0$ (and therefore $g'(t) \neq 0$).

Note. When $f'(t) = g'(t) = 0$, for curves C with additional properties, natural definition(s) of tangent(s) do exist but are beyond Calc II.

Example

Find the tangent to the curve

$$\gamma : \begin{cases} x = \cos t \\ y = \sin t \end{cases}, t \in [0, 2\pi) \text{ at } t = \frac{\pi}{4}, t = \frac{2\pi}{3}, t = \pi.$$

Recall C : $\begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a, b]$, tangent vector at t is $(x'(t), y'(t))$.

We write informally $x = x(t), y = y(t)$ to simplify notation.

- Suppose we could eliminate the parameter t and write $y = F(x)$ for some function F near the point $(x, y) = (x(t), y(t))$.
- Suppose in $x'(t) \neq 0$ for some t .

$$\begin{array}{lcl}
 y & = & F(x) \\
 \frac{dy}{dt} & = & \frac{d}{dt}(F(x)) \\
 & = & \frac{dF}{dx} \frac{dx}{dt} = \frac{dy}{dx} \frac{dx}{dt} \\
 \frac{dy}{dx} & = & \frac{\frac{dy}{dt}}{\frac{dx}{dt}}
 \end{array}
 \quad \left| \begin{array}{l} \text{apply } \frac{d}{dt} \\ \text{use chain rule} \\ \text{divide by } x'(t) \end{array} \right.$$

Observation

If $\frac{dx}{dt} \neq 0$, we have $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

Example

A curve C is defined by $x = t^2$, $y = t^3 - 3t$.

- 1 Show C traverses $(x, y) = (3, 0)$ for two values of t ; find the tangent slopes for both of these values.
- 2 Find the points on C where the tangents are horizontal or vertical.
- 3 Find two intervals where we can write y as a function of x .
- 4 Determine concavity intervals of the functions found in item 3.

Example

A curve C is defined by $x = t^2, y = t^3 - 3t$.

- 1 Show C traverses $(x, y) = (3, 0)$ for two values of t ; find the tangent slopes for both of these values.
 - $3 = x = t^2$ if $t = \pm\sqrt{3}$.
 - $0 = y = t^3 - 3t = t(t^2 - 3)$ if $t = 0$ or $\pm\sqrt{3}$.
 - Therefore the point $(3, 0)$ is traversed when t equals $\sqrt{3}$ or $-\sqrt{3}$.
 - $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2-3}{2t}$.
 - Plug in $t = \pm\sqrt{3}$: $\frac{dy}{dx}|_{t=\pm\sqrt{3}} = \frac{3(\pm\sqrt{3})^2 - 3}{2(\pm\sqrt{3})} = \pm \frac{6}{2\sqrt{3}} = \pm\sqrt{3}$
- Therefore the tangents at $(3, 0)$ have slopes $\pm\sqrt{3}$.

Example

A curve C is defined by $x = t^2, y = t^3 - 3t$.

- ② Find the points on C where the tangents are horizontal or vertical.

Horizontal tangent:

$$\frac{dy}{dt} = 0$$

$$3t^2 - 3 = 0$$

$$3(t^2 - 1) = 0$$

$$t = \pm 1$$

$\frac{dx}{dt} \neq 0$ when $t = \pm 1$, so there are horizontal tangents when $t = \pm 1$.

The points are $(1, 2)$ and $(1, -2)$.

Vertical tangent:

$$\frac{dx}{dt} = 0$$

$$2t = 0$$

$$t = 0$$

$\frac{dy}{dt} \neq 0$ when $t = 0$, so there is a vertical tangent when $t = 0$.

The point is $(0, 0)$.

Example

A curve C is defined by $x = t^2, y = t^3 - 3t$.

- ③ Find two intervals where we can write y as a function of x .

From $x = t^2$ we have that $t = \pm\sqrt{x}$. Therefore, when $t > 0$, we have that $t = \sqrt{x}$. Since that determines uniquely t via x , this means that for $t > 0$ y is a function of x . In other words, for $t > 0$, the curve satisfies the vertical line test. Similarly we conclude that when $t < 0$, y is a function of x .

Example

A curve C is defined by $x = t^2, y = t^3 - 3t$.

- 4 Determine the concavity intervals of the functions found in item 3.

Find the second derivative:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} \left(\frac{3t^2-3}{2t} \right)}{2t} \\ &= \frac{\frac{d}{dt} \left(\frac{3}{2} \left(t - \frac{1}{t} \right) \right)}{2t} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t} \\ &= \frac{\frac{3t^2+3}{2t^2}}{2t} = \frac{3(t^2+1)}{4t^3}\end{aligned}$$

Therefore y as a function of x (which is a function of t) is concave up when $t > 0$ and concave down when $t < 0$.

Example

Consider the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$.

- 1 At what points is the tangent horizontal?
- 2 At what points is the tangent vertical?

Example

Consider the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$.

① At what points is the tangent horizontal?

- The slope of the tangent is $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{r \sin \theta}{r(1 - \cos \theta)} = \frac{\sin \theta}{1 - \cos \theta}$
- The tangent is horizontal when $dy/dx = 0$, that is, when $dy/d\theta = 0$ and $dx/d\theta \neq 0$.
- $r \sin \theta = dy/d\theta = 0$ if $\theta = n\pi$, where n is any integer.
- $r(1 - \cos \theta) = dx/d\theta = 0$ if $\theta = 2n\pi$, where n is any integer.
- Therefore there is a horizontal tangent when $\theta = (2n + 1)\pi$.

Example

Consider the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$.

② At what points is the tangent vertical?

- When $\theta = 2n\pi$ both $dy/d\theta$ and $dx/d\theta$ are 0.
- To see if there is a vertical tangent, use L'Hospital's Rule.

$$\lim_{\theta \rightarrow 2n\pi^+} \frac{dy}{dx} = \lim_{\theta \rightarrow 2n\pi^+} \frac{\sin \theta}{1 - \cos \theta} = \lim_{\theta \rightarrow 2n\pi^+} \frac{\cos \theta \rightarrow 1}{\sin \theta \rightarrow 0^+}$$

- Therefore $\lim_{\theta \rightarrow 2n\pi^+} (dy/dx) = \infty$.
- A similar argument shows $\lim_{\theta \rightarrow 2n\pi^-} (dy/dx) = -\infty$.
- Therefore there is a vertical tangent when $\theta = 2n\pi$.

Tangents to Polar Curves

To find the tangent line to a polar curve $r = f(\theta)$, regard θ as a parameter and write the parametric equations as

$$x = r \cos \theta = f(\theta) \cos \theta \quad y = r \sin \theta = f(\theta) \sin \theta$$

Then use the formula for the slope of a parametric curve:

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ &= \frac{\frac{d}{d\theta} (f(\theta) \sin \theta)}{\frac{d}{d\theta} (f(\theta) \cos \theta)} \\ &= \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta + f(\theta)(-\sin \theta)} \\ &= \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} \end{aligned}$$

Example

Find the points on $r = 1 + \sin \theta$ where the tangent is horizontal or vertical.

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = \frac{\cos \theta \sin \theta + (1 + \sin \theta) \cos \theta}{\cos \theta \cos \theta - (1 + \sin \theta) \sin \theta} \\ &= \frac{\cos \theta (1 + 2 \sin \theta)}{1 - 2 \sin^2 \theta - \sin \theta} = \frac{\cos \theta (1 + 2 \sin \theta)}{(1 + \sin \theta)(1 - 2 \sin \theta)}\end{aligned}$$

- $\cos \theta (1 + 2 \sin \theta) = 0$
when $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.
- $(1 + \sin \theta)(1 - 2 \sin \theta) = 0$
when $\theta = \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$.

- Horizontal tangents at $(2, \pi/2)$, $(1/2, 7\pi/6)$, and $(1/2, 11\pi/6)$.
- Vertical tangents at $(3/2, \pi/6)$, and $(3/2, 5\pi/6)$.
- If $\theta = 3\pi/2$, top and bottom are both 0, so use L'Hospital's Rule.

$$\lim_{\theta \rightarrow 3\pi/2^-} \frac{dy}{dx} = \lim_{\theta \rightarrow 3\pi/2^-} \frac{1 + 2 \sin \theta}{1 - 2 \sin \theta} \cdot \lim_{\theta \rightarrow 3\pi/2^-} \frac{\cos \theta}{1 + \sin \theta} = -\frac{1}{3} \lim_{\theta \rightarrow 3\pi/2^-} \frac{-\sin \theta}{\cos \theta} = \infty$$

Arc Length

- What do we mean by the length of a curve?
- The length of a polygon is easy to compute: add up the length of the line segments that form the polygon.
- If the curve is a circle, approximate it by a polygon.
- Then take the limit as the number of segments of the polygon goes to ∞ .

Let γ be the curve $\gamma : \begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a, b]$

- Divide $[a, b]$ into n subintervals with endpoints $t_0, t_1, \dots, t_n$ and equal width Δt .
- The points $P_i = (x(t_i), y(t_i))$ lie on the curve γ . The lengths of the segments with endpoints with consecutive indices from $P_0, P_1, \dots, P_n$ approximate the length of the curve γ .
- The length L of the curve γ is the limit of the lengths of these segments as $n \rightarrow \infty$.

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1} P_i|$$

Let γ be the curve $\gamma : \begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a, b]$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i| = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(x'(s_i))^2 + (y'(r_i))^2} \Delta t \\ &= \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt \end{aligned}$$

- If f has continuous derivative, we can compute the above limit.
- Let $\begin{cases} x_i = x(t_i) \\ y_i = y(t_i) \end{cases}$, and $\begin{cases} \Delta x = x_i - x_{i-1} = x(t_i) - x(t_{i-1}) \\ \Delta y = y_i - y_{i-1} = y(t_i) - y(t_{i-1}) \end{cases}$.
- Then $|P_iP_{i-1}| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$.
- Mean Value Theorem: there exist numbers s_i and r_i between t_{i-1} and t_i such that $x(t_i) - x(t_{i-1}) = x'(s_i)(t_i - t_{i-1})$ and $y(t_i) - y(t_{i-1}) = y'(r_i)(t_i - t_{i-1})$.
- $\Delta x = x'(s_i)\Delta t$, $\Delta y = y'(r_i)\Delta t$.

$$\begin{aligned} |P_{i-1}P_i| &= \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(x'(s_i)\Delta t)^2 + (y'(r_i)\Delta t)^2} \\ &= \sqrt{(x'(s_i))^2 + (y'(r_i))^2} \sqrt{(\Delta t)^2} = \sqrt{(x'(s_i))^2 + (y'(r_i))^2} \Delta t \end{aligned}$$

The Arc Length Formula

Let $\gamma : \begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a, b]$.

Definition

Suppose $x'(t)$ and $y'(t)$ (exist and) are continuous on $[a, b]$. Then the length of the curve γ is defined as

$$\begin{aligned} L(\gamma) &= \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad \text{in Leibniz notation .} \end{aligned}$$

Arc length of graph of a function

Question

What is the length of the graph of the curve given by the graph of $y = f(x)$?

- The graph of $y = f(x)$ is written as a curve as

$$\gamma : \begin{cases} x = t \\ y = f(t) \end{cases}, t \in [a, b] \quad .$$

- In other words, the question asks what is the length $L(\gamma)$ of γ . That is a straightforward computation:

$$L(\gamma) = \int \sqrt{(x'(t))^2 + (y'(t))^2} dt = \int \sqrt{1 + (f'(t))^2} dt$$

The Arc Length Formula

Definition

Suppose f' exists and is continuous on $[a, b]$. Then the length of the curve $y = f(x)$, $a \leq x \leq b$, is

$$\begin{aligned} L &= \int_a^b \sqrt{1 + (f'(x))^2} \, dx \\ &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx \quad (\text{in Leibniz notation}) . \end{aligned}$$

Example

Find the length of the arc of $y^2 = x^3$ between $(1, 1)$ and $(4, 8)$.

- For the top half of the curve we have:
- $y = x^{3/2}$ and $y' = \frac{3}{2}x^{1/2}$.
- $u = 1 + \frac{9}{4}x$ and $du = \frac{9}{4}dx$.
- When $x = 1$, $u = \frac{13}{4}$.
- When $x = 4$, $u = 10$.

$$\begin{aligned} L &= \int_1^4 \sqrt{1 + (y')^2} dx \\ &= \int_1^4 \sqrt{1 + \frac{9}{4}x} dx = \int_{13/4}^{10} \frac{4}{9} \sqrt{u} du \\ &= \frac{4}{9} \left[\frac{2}{3} u^{3/2} \right]_{13/4}^{10} = \frac{8}{27} \left(10^{3/2} - \left(\frac{13}{4} \right)^{3/2} \right) \end{aligned}$$

Example

Find the length of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

$$\begin{aligned}
 L &= \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{x=0}^{x=1} \sqrt{1 + 4x^2} dx \quad \left| \begin{array}{l} \frac{dy}{dx} = 2x \\ \text{Set } x = \frac{1}{2} \tan \theta \end{array} \right. \\
 &= \int_{\theta=0}^{\theta=\arctan 2} \sqrt{1 + \tan^2 \theta} d\left(\frac{1}{2} \tan \theta\right) \\
 &= \int_{\theta=0}^{\theta=\arctan 2} \sec \theta \cdot \frac{1}{2} \sec^2 \theta d\theta = \frac{1}{2} \int_0^{\alpha} \sec^3 \theta d\theta \quad \left| \begin{array}{l} \text{Set } \alpha = \arctan 2 \end{array} \right. \\
 &= \frac{1}{2} \cdot \left[\frac{1}{2} (\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|) \right]_{\theta=0}^{\theta=\alpha} \\
 &= \frac{1}{4} (\sec \alpha \tan \alpha + \ln |\sec \alpha + \tan \alpha|) \\
 &= \frac{1}{4} (2 \cdot \sqrt{5} + \ln |\sqrt{5} + 2|)
 \end{aligned}$$

Example

Find the length of the curve γ .

$$\gamma : \begin{cases} x(t) = \sqrt{t} - 2t \\ y(t) = \frac{8}{3}t^{\frac{3}{4}} \end{cases}, t \in [1, 4]$$

We have that $x'(t) = \frac{1}{2\sqrt{t}} - 2$ and $y'(t) = \frac{8}{3} \cdot \frac{3}{4}t^{-\frac{1}{4}} = 2t^{-\frac{1}{4}}$.

$$\begin{aligned} L(\gamma) &= \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt = \int_1^4 \sqrt{\left(\frac{1}{2\sqrt{t}} - 2\right)^2 + \left(2t^{-\frac{1}{4}}\right)^2} dt \\ &= \int_1^4 \sqrt{\frac{1}{4t} - \frac{2}{\sqrt{t}} + 4 + \frac{4}{\sqrt{t}}} dt \\ &= \int_1^4 \sqrt{\frac{1}{4t} + \frac{2}{\sqrt{t}} + 4} dt = \int_1^4 \sqrt{\left(\frac{1}{2\sqrt{t}} + 2\right)^2} dt \\ &= \int_1^4 \left(\frac{1}{2\sqrt{t}} + 2\right) dt = \left[\sqrt{t} + 2t\right]_1^4 = \sqrt{4} + 2 \cdot 4 - (\sqrt{1} + 2 \cdot 1) = 7. \end{aligned}$$

Example $((a + b)^2, (a - b)^2, 2ab = 1/2)$

Find the length of the arc of $y = \frac{1}{6}e^{3x} + \frac{1}{6}e^{-3x}$ from $x = 0$ to $x = 1$.

$$y' = \frac{1}{2}e^{3x} - \frac{1}{2}e^{-3x}.$$

$$\begin{aligned}(y')^2 &= \frac{1}{4}e^{6x} - \frac{1}{4}e^{3x}e^{-3x} - \frac{1}{4}e^{3x}e^{-3x} + \frac{1}{4}e^{-6x} \\ &= \frac{1}{4}e^{6x} - \frac{1}{2} + \frac{1}{4}e^{-6x}.\end{aligned}$$

$$\begin{aligned}L &= \int_0^1 \sqrt{1 + (y')^2} dx = \int_0^1 \sqrt{1 + \frac{1}{4}e^{6x} - \frac{1}{2} + \frac{1}{4}e^{-6x}} dx \\ &= \int_0^1 \sqrt{\frac{1}{4}e^{6x} + \frac{1}{2} + \frac{1}{4}e^{-6x}} dx = \int_0^1 \sqrt{\left(\frac{1}{2}e^{3x} + \frac{1}{2}e^{-3x}\right)^2} dx \\ &= \int_0^1 \left(\frac{1}{2}e^{3x} + \frac{1}{2}e^{-3x}\right) dx = \left[\frac{1}{6}e^{3x} - \frac{1}{6}e^{-3x}\right]_0^1 = \frac{e^3 - e^{-3}}{6}.\end{aligned}$$

Example

Find the length of one arch of the cycloid

$$x = r(\theta - \sin \theta), \quad y = r(1 - \cos \theta).$$

The first arch is $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(r(1 - \cos \theta))^2 + (r \sin \theta)^2} d\theta \\ &= \int_0^{2\pi} \sqrt{r^2(1 - 2\cos \theta + \cos^2 \theta + \sin^2 \theta)} d\theta = r \int_0^{2\pi} \sqrt{2(1 - \cos \theta)} d\theta \end{aligned}$$

Use the identity $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$. Then

$$\sqrt{2(1 - \cos \theta)} = \sqrt{4 \sin^2(\theta/2)} = 2|\sin(\theta/2)| = 2 \sin(\theta/2)$$

$$L = r \int_0^{2\pi} 2 \sin(\theta/2) d\theta = r [-4 \cos(\theta/2)]_0^{2\pi} = 8r$$

Arc Length

To find the arc length of a polar curve $r = f(\theta)$, $a \leq \theta \leq b$, regard θ as a parameter. Then the derivatives of the parametric equations are

$$\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta \quad \frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta$$

and

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= \left(\frac{dr}{d\theta}\right)^2 \cos^2 \theta - 2r \frac{dr}{d\theta} \cos \theta \sin \theta + r^2 \sin^2 \theta \\ &\quad + \left(\frac{dr}{d\theta}\right)^2 \sin^2 \theta + 2r \frac{dr}{d\theta} \sin \theta \cos \theta + r^2 \cos^2 \theta \\ &= \left(\frac{dr}{d\theta}\right)^2 + r^2 \end{aligned}$$

The arc length is

$$L = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Example

Find the length of the cardioid $r = 1 + \sin \theta$. The full length is given by $0 \leq \theta \leq 2\pi$.

$$\begin{aligned}
 L &= \int_0^{2\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(1 + \sin \theta)^2 + \cos^2 \theta} d\theta \\
 &= \int_0^{2\pi} \sqrt{2 + 2 \sin \theta} \frac{\sqrt{2 - 2 \sin \theta}}{\sqrt{2 - 2 \sin \theta}} d\theta = \int_0^{2\pi} \frac{\sqrt{4 - 4 \sin^2 \theta}}{\sqrt{2 - 2 \sin \theta}} d\theta \\
 &= \int_0^{2\pi} \frac{\sqrt{4 \cos^2 \theta}}{\sqrt{2 - 2 \sin \theta}} d\theta = \int_0^{2\pi} \frac{2|\cos \theta|}{\sqrt{2 - 2 \sin \theta}} d\theta \\
 &= \int_0^{\pi/2} \frac{2 \cos \theta}{\sqrt{2 - 2 \sin \theta}} d\theta + \int_{\pi/2}^{3\pi/2} \frac{-2 \cos \theta}{\sqrt{2 - 2 \sin \theta}} d\theta + \int_{3\pi/2}^{2\pi} \frac{2 \cos \theta}{\sqrt{2 - 2 \sin \theta}} d\theta \\
 &= \left[-2\sqrt{2 - 2 \sin \theta}\right]_0^{\pi/2} + \left[2\sqrt{2 - 2 \sin \theta}\right]_{\pi/2}^{3\pi/2} + \left[-2\sqrt{2 - 2 \sin \theta}\right]_{3\pi/2}^{2\pi} \\
 &= -2(0 - \sqrt{2}) + 2(2 - 0) - 2(\sqrt{2} - 2) = 8
 \end{aligned}$$